PROVISIONAL PATENT SPECIFICATION

Method and System for Geometric Fault Localization in Photonic Computing Meshes Using Gaussian Curvature of the Coupled-Mode Constraint Surface

Applicant: William Lloyd

Address: Coffs Harbour, NSW, Australia

Date: April 2026

1 TECHNICAL FIELD

This invention relates to methods and systems for fault diagnosis and quality screening in silicon photonic integrated circuits. More particularly, the invention provides a method for localizing splitting ratio faults in individual Mach-Zehnder interferometers (MZIs) within cascaded photonic computing meshes by computing the Gaussian curvature of the coupled-mode theory constraint surface for each component, wherein the vanishing of the Gaussian curvature at the design point constitutes a symmetry-derived health criterion that is independent of calibrated thresholds, and wherein the diagnostic is exactly invariant under perturbations of the measured output power due to the linear (Class 1) participation of the output variable in the constraint equation.

The invention extends and applies the diagnostic methods described in Australian Provisional Patent Application No. AU 2026902758 (the “first parent application”), filed 28 March 2026, and Australian Provisional Patent Application No. AU 2026902926 (the “second parent application”), filed April 2026, both by the same applicant.

2 BACKGROUND

2.1 The Scaling Wall in Photonic Computing

Silicon photonic computing performs matrix-vector multiplication by routing light through meshes of Mach-Zehnder interferometers, each built from two directional couplers and a tunable phase shifter. The phase shifter compensates for fabrication imperfections in optical path length. The directional coupler’s splitting ratio, however, is frozen at fabrication and depends on the waveguide gap with exponential sensitivity: $\kappa \propto \exp(-\gamma d)$, where a ±10 nm gap variation causes a 15–30% change in the coupling coefficient.

Hamerly, Bandyopadhyay, and Englund (Nature Communications, 2022) proved that the residual matrix error after self-configuration scales as $\varepsilon\_c = (2/3)N\sigma^2$, where N is the mesh dimension and σ is the splitting ratio standard deviation. The fraction of achievable unitary transformations collapses exponentially as $\exp(-N^3\sigma^2/3)$. At $\sigma = 5\%$ (typical for broadband couplers), practical computation degrades beyond approximately $N = 64$.

2.2 Limitation of Existing Approaches

The field's response has been architectural: fault-tolerant MZI designs using three beam splitters instead of two, achieving asymptotic fault tolerance but requiring 1.5–2× more components — increasing the very fabrication variability they are designed to tolerate.

The alternative — identifying and correcting individual faulty components — has been abandoned because fault localization in large meshes is an ill-posed inverse problem: multiple distinct fault patterns produce indistinguishable output signatures. Existing diagnostic methods (CLIPP non-invasive monitoring, self-calibration protocols) measure aggregate performance but cannot attribute degradation to individual MZIs within a cascaded mesh.

No published work uses geometric or topological invariants as compact diagnostic signatures for individual photonic circuit components.

2.3 Problem Statement

There exists a need for a fault localization method for photonic computing meshes that:

(a) Identifies which individual MZI in a cascaded mesh has a splitting ratio fault, not merely that a fault exists somewhere in the mesh;

(b) Operates from a theoretically grounded health criterion rather than a calibrated empirical threshold;

(c) Is robust to imperfect isolation of adjacent components during testing;

(d) Detects errors at the threshold where they begin to affect computational performance;

(e) Runs in sub-millisecond time per component, compatible with wafer-level screening and in-situ monitoring;

(f) Requires no training data, no electromagnetic simulation, and no internal optical probes.

3 SUMMARY OF THE INVENTION

The present invention provides a method for localizing splitting ratio faults in individual Mach-Zehnder interferometers within a photonic computing mesh by computing the Gaussian curvature K_G of the coupled-mode theory constraint surface for each MZI at its operating point.

The method exploits a symmetry property of the MZI constraint equation: at the design splitting ratio of $r = 0.5$, the derivative of the splitting function $4r(1-r)$ vanishes, making the constraint surface locally cylindrical in the splitting ratio direction and yielding $K\_G = 0$. For any splitting ratio $r \neq 0.5$, this symmetry is broken and $K\_G \neq 0$, with magnitude proportional to $|r - 0.5|^2$. This $K\_G = 0$ condition is the unique geometric signature of a perfect 50:50 beam splitter — a property of the constraint equation's structure, not a calibrated threshold.

The method further exploits the Class 1 coupling structure of the MZI constraint: the measured output power T enters the constraint equation linearly, meaning all second partial derivatives involving T vanish identically. Consequently, K_G is exactly invariant under perturbations of T, making the diagnostic immune to imperfect isolation of adjacent MZIs during testing. This exact invariance is a structural consequence of the constraint geometry, not an approximation.

The invention applies the intrinsic geometric diagnostic method of the second parent application (AU 2026902926) to the specific domain of silicon photonic computing, and inherits the parameterisation independence, coupling class taxonomy, and cross-domain comparison capabilities described therein.

4 DETAILED DESCRIPTION

4.1 The Constraint Surface for a Directional Coupler

The coupled-mode theory constraint equation for a directional coupler is:

$$F(\kappa, \Delta\beta, R) = R - [\kappa^2/(\kappa^2+\Delta\beta^2)] \cdot \sin^2(\sqrt{(\kappa^2+\Delta\beta^2)} \cdot L) = 0$$

where κ is the coupling coefficient (μm^{-1}), $\Delta\beta$ is the propagation constant mismatch (μm^{-1}), R is the power transfer ratio, and L is the interaction length (fixed per device). This equation defines a two-dimensional constraint surface in the three-dimensional variable space (κ, $\Delta\beta$, R), encoding the complete coupling geometry of the device.

4.2 The MZI Constraint and Symmetry Argument

For a symmetric MZI with splitting ratio r, the constraint simplifies to:

$$F(r, \theta, T) = T - 4r(1-r) \cdot \sin^2(\theta/2) = 0$$

where θ is the differential phase and T is the cross-port power transfer.

The Gaussian curvature K_G is computed from the bordered Hessian of F as described in the second parent application (AU 2026902926, Method 1, Section 4.2.2):

$$K\_G = \det(H^*) / |\nabla F|^4$$

At $r = 0.5$ exactly, the following conditions hold:

(i) The gradient component $\partial F/\partial r = -4(1-2r) \cdot \sin^2(\theta/2)$ vanishes because $1-2r = 0$. The constraint surface becomes locally cylindrical in the r-direction.

(ii) The bordered Hessian determinant vanishes in the r-dependent sector, yielding K_G = 0.

(iii) For any $r \neq 0.5$, the derivative $4(1-2r) \neq 0$, the gradient acquires an r-component, and K_G $\neq 0$.

Therefore K_G = 0 is the unique geometric signature of a perfect 50:50 beam splitter. The magnitude of the curvature is proportional to $|r - 0.5|^2$, forming a parabolic calibration curve that encodes both the presence and severity of splitting ratio error.

4.3 Exact Invariance Under Output Perturbation

In the MZI constraint $F(r, \theta, T) = T - 4r(1-r)\sin^2(\theta/2)$, the measured output T enters linearly. It is the Class 1 passenger in the coupling class taxonomy of the second parent application. The following second partial derivatives vanish identically:

$$\partial^2F/\partial T^2 = 0, \quad \partial^2F/\partial T\partial r = 0, \quad \partial^2F/\partial T\partial\theta = 0$$

Because K_G is computed entirely from second partial derivatives of F (via the bordered Hessian), and all second derivatives involving T vanish, K_G is exactly invariant under perturbations of T. This has the following operational consequence: imperfect isolation of adjacent MZIs during testing perturbs the measured output power T, but this perturbation does not affect the computed K_G. The diagnostic is structurally immune to isolation errors, not approximately immune.

Numerical verification confirms this: with residual coupling of 0.25% from a neighbouring MZI with 0.1 rad phase offset, the K_G perturbation is 1.3×10^{-13} — thirteen orders of magnitude below any detection threshold.

4.4 Role Transposition Invariance

By Theorem 8.1 of the Directive-Based Products Framework (referenced in the second parent application), K_G is invariant under all six permutations of the variable role assignment. Whether the test station measures the MZI by sweeping phase and inferring splitting ratio, or by fixing phase and varying input power, or by measuring cross-port transmission and back-calculating coupling, the K_G value is identical. This eliminates test-setup dependence from the manufacturing screen: different factory configurations produce the same geometric fingerprint for the same component.

4.5 Fault Localization Protocol

4.5.1 Isolation Protocol

For a mesh of N_MZI Mach-Zehnder interferometers, the fault localization proceeds as follows:

(a) For each MZI in the mesh: set all other MZIs to pass-through ($\theta = 0$). Set the test MZI to an operating phase (e.g., $\theta = \pi/2$).

(b) Measure the cross-port power transfer T at the test MZI's output.

(c) Compute K_G from the constraint equation $F(r, \theta, T) = 0$ at the measured operating point, using the bordered Hessian formula.

(d) Compare the computed K_G against the healthy baseline. A deviation exceeding a threshold flags the component as faulty.

The healthy baseline is K_G ≈ 0 (with a small systematic offset from the vanishing gradient at $r = 0.5$). The detection threshold is determined by the measurement noise floor and the minimum splitting ratio error of concern.

4.5.2 Multiple Simultaneous Faults

Because K_G is computed independently for each MZI, the diagnostic signatures of different faulty components do not interfere. Multiple simultaneous faults are localized without ambiguity. This is a direct consequence of the per-component geometric computation — no inverse problem is being solved, so there is no ill-posedness.

4.5.3 Detection Sensitivity and Scaling Law Connection

The minimum detectable splitting ratio error is determined by the measurement noise floor relative to the parabolic K_G calibration curve. At typical photonics test equipment precision (≤1% measurement noise), the detection threshold is approximately 4% ($\Delta r = 0.02$).

This threshold connects directly to the Hamerly et al. scaling law. If K_G screening removes all components with $|\Delta r| > 0.02$, the effective σ of the remaining truncated distribution drops from the population value (e.g., 0.05) to approximately 0.019, reducing the residual matrix error ε_c by 7×. Equivalently, the same error budget that previously constrained the mesh to N ≈ 64 now supports N ≈ 450. The detection threshold corresponds to the geometric inflection point where individual component errors begin to dominate the mesh error budget.

4.5.4 Computation Time

The bordered Hessian computation for each MZI requires evaluation of partial derivatives of F (which are analytic for the coupled-mode constraint) and a 4×4 determinant. For a mesh of N_MZI components, the total computation scales linearly with N_MZI. Demonstrated performance: 6 MZIs diagnosed in 0.04 seconds; 2,352 couplers diagnosed in 1.3 seconds (1,860 couplers/second); estimated 1.1 seconds for a 64×64 mesh (~2,048 MZIs). This is compatible with real-time monitoring between mesh reconfigurations and with wafer-level screening during manufacturing.

4.6 Extension to General Directional Coupler Diagnostics

The method extends beyond the symmetric MZI to the full coupled-mode theory constraint surface $F(\kappa, \Delta\beta, R) = 0$, enabling diagnosis of:

(a) Gap errors: detected via K_G deviation from baseline, with geometric sensitivity proportional to $\exp(-2\gamma d)$ (exponentially higher at small gaps).

(b) Cladding voids: detected via simultaneous perturbation of κ (reduced) and Δβ (increased), producing a distinctive K_G signature.

(c) Thermal drift: detected via Δβ perturbation from transverse thermal gradients, producing a K_G shift proportional to the temperature gradient magnitude.

(d) Combined faults: multiple simultaneous fault mechanisms produce composite K_G deviations that are additive in the curvature domain.

4.7 Cross-Domain Validation

The parameterisation independence of K_G (Gauss's Theorema Egregium) enables validation of the diagnostic against unrelated physical systems using the cross-domain equivalence method of the second parent application (AU 2026902926, Method 4). If K_G correctly identifies geometric equivalences and differences across domains, the diagnostic is measuring genuine constraint geometry rather than domain-specific artifacts.

The photonic directional coupler constraint surface is geometrically equivalent to quantum Rabi oscillations (RMS = 0.016), confirming that both are SU(2) rotations. It is structurally different from Van der Waals criticality (RMS > 1.0) and ideal gas coupling (RMS = 0.274), confirming discriminative power.

4.8 Constraint Surface Characterisation

The K_G diagnostic profile characterises the full range of coupler physics:

(a) Synchronous coupler: K_G = 2.5×10^{-4} (flat baseline, Class 1). The near-zero K_G confirms the locally cylindrical geometry at the design point.

(b) Asynchronous coupler: K_G = −336.5 (saddle geometry, Class 1). The K_G $\propto 1/\Delta\beta^2$ relationship directly quantifies fabrication sensitivity as a geometric invariant.

(c) Kerr nonlinear coupler: K_G = 0.030 (bowl geometry, Class 1). The closure concentration ratio of 263 indicates that single-point K_G captures only 0.4% of total geometric content; the recursive closure pipeline of the second parent application is essential.

(d) Lossy coupler: K_G = 1.958 (bowl geometry, Class 1). Loss-stable diagnostic.

(e) Ring resonator at critical coupling: K_G = 1.388 (bowl geometry, Class 2). The K_G spike to −418.7 at the critical coupling condition ($a = t$) geometrically identifies the singularity and the principal escape direction.

All Class 1 systems transition to Class 2 under recursive closure (Method 2 of the second parent application), confirming the universal class transition property.

5 WORKED EXAMPLES

5.1 Example 1: Single Fault Localization in a 4×4 Clements Mesh

System: A 4×4 Clements mesh comprising 6 MZIs arranged in 4 layers. MZI_4 has a splitting ratio fault of $r = 0.55$ (design value 0.50). All other MZIs have $r = 0.50$.

Step 1: Set all MZIs except MZI_0 to pass-through ($\theta = 0$). Set MZI_0 to $\theta = \pi/2$. Measure T. Compute K_G = −0.00103 (healthy baseline). **Result: MZI_0 cleared.**

Step 2: Repeat for MZI_1 through MZI_3. All return K_G ≈ −0.00103. **Result: MZI_1, MZI_2, MZI_3 cleared.**

Step 3: Test MZI_4. Compute K_G = 0.02328 (deviation of 0.02431 from baseline, exceeding threshold of 0.002). **Result: MZI_4 flagged as faulty.**

Step 4: Test MZI_5. Compute K_G ≈ −0.00103. **Result: MZI_5 cleared.**

Localization accuracy: 6/6 correct. The faulty MZI is identified and all healthy MZIs are cleared. Total computation time: 0.04 seconds.

5.2 Example 2: Multiple Simultaneous Faults

System: Same 4×4 Clements mesh. MZI_1 has $r = 0.54$, MZI_4 has $r = 0.47$. All other MZIs have $r = 0.50$.

The isolation protocol tests each MZI individually. MZI_1 returns K_G deviating from baseline; MZI_4 returns K_G deviating from baseline; all other MZIs return K_G consistent with baseline. Both faults are correctly localized, all healthy MZIs are correctly cleared. Score: 6/6 correct. The K_G signatures do not interfere because the diagnostic is computed independently for each component.

5.3 Example 3: Exact Invariance Under Imperfect Isolation

System: Same 4×4 Clements mesh. MZI_0 is under test. A neighbouring MZI has a residual phase offset of 0.1 rad (imperfect pass-through), contributing 0.25% residual coupling to the measured output T.

The perturbation in T shifts the measured cross-port power by approximately 0.0025. However, because $\partial^2F/\partial T^2 = \partial^2F/\partial T\partial r = \partial^2F/\partial T\partial\theta = 0$ (T enters linearly), the computed K_G changes by 1.3×10^{-13}. The diagnostic result is unaffected. The exact invariance under T perturbation is confirmed.

5.4 Example 4: Scaling Law Impact

A photonic foundry produces MZIs with splitting ratio standard deviation $\sigma = 0.05$. Using the Hamerly scaling law $\varepsilon\_c = (2/3)N\sigma^2$:

Without screening: At $N = 64$, $\varepsilon\_c = 0.107$. The maximum practical mesh dimension for this error budget is $N \approx 64$.

With K_G screening: All components with $|\Delta r| > 0.02$ are identified and removed. The effective σ of the screened population drops to approximately 0.019. At this σ, the same error budget ($\varepsilon\_c = 0.107$) supports $N \approx 450$ — a 7× improvement in the scaling ceiling.

6 SYSTEM EMBODIMENT

The method of the invention may be implemented as an extension to the diagnostic system described in the parent applications, comprising:

(a) A photonic constraint module that encodes the coupled-mode theory constraint equation $F(\kappa, \Delta\beta, R) = 0$ and the MZI constraint equation $F(r, \theta, T) = 0$, computes their partial derivatives analytically, and evaluates the bordered Hessian at measured operating points.

(b) A mesh localization module that implements the isolation protocol, sequentially testing each MZI in a cascaded mesh and computing K_G for each component.

(c) A fault classification module that compares each component's K_G against the healthy baseline, flags deviations exceeding a noise-floor-derived threshold, and reports the inferred splitting ratio error via the parabolic calibration curve.

(d) A screening integration module that interfaces with wafer-level test equipment to provide known-good-die screening based on per-component K_G signatures, outputting accept/reject decisions and per-component quality scores for each mesh on a wafer.

(e) A cross-domain validation module that confirms the diagnostic is measuring genuine constraint geometry by comparing photonic K_G profiles against profiles from unrelated physical systems, using the cross-domain equivalence method of the second parent application.

These modules may operate independently or in combination with the intrinsic geometric diagnostic, recursive closure, centrifuge, and cross-domain equivalence methods of the second parent application. The photonic constraint module (a) and mesh localization module (b) may be used as a standalone manufacturing screen without requiring the full framework.

In a further embodiment, the system operates in continuous monitoring mode during photonic mesh operation, periodically setting MZIs to pass-through during maintenance windows and computing K_G to detect progressive splitting ratio drift before computational performance degrades.

In a further embodiment, the system provides real-time monitoring integrated with mesh reconfiguration protocols, triggering re-routing around degraded components when K_G exceeds the fault threshold during operation.

7 SCOPE OF APPLICATION

The method and system of the invention are applicable to, but not limited to:

Silicon photonic computing meshes (MZI-based matrix-vector multipliers, Clements and Reck architectures, programmable photonic processors);

Photonic integrated circuit manufacturing (wafer-level screening, known-good-die selection, foundry quality control, process characterisation);

Directional coupler characterisation (splitting ratio measurement, gap error quantification, cladding defect detection, thermal sensitivity profiling);

Ring resonator diagnostics (critical coupling identification, loss characterisation, singularity detection);

Nonlinear photonic devices (Kerr coupler bifurcation mapping, self-phase modulation diagnostics);

Any photonic device whose coupling physics can be expressed as a three-variable constraint equation $F(x_1, x_2, x_3) = 0$.

8 ADVANTAGES OVER PRIOR ART

1. Per-component localization: The method identifies which individual MZI in a cascaded mesh is faulty, solving the inverse problem that existing methods have abandoned. No other published method provides per-component geometric fault signatures for photonic circuits.

2. Symmetry-derived health criterion: The K_G = 0 condition at r = 0.5 is a mathematical property of the constraint equation, not a calibrated empirical threshold. It does not drift, does not require recalibration, and does not depend on measurement equipment or environmental conditions.

3. Exact isolation invariance: The Class 1 structure of the MZI constraint makes K_G exactly invariant under perturbations of the measured output. Imperfect isolation of adjacent MZIs during testing does not degrade the diagnostic. This is a structural property, not an approximation.

4. Test-setup independence: The role transposition invariance (Theorem 8.1 of the parent framework) guarantees that the K_G value is identical regardless of which variable is treated as the driving input, the medium, or the output. Different factory test configurations produce the same geometric fingerprint.

5. Scaling wall reduction: Screening at the 4% detection threshold reduces the effective population variance by 7×, raising the practical scaling ceiling from $N \approx 64$ to $N \approx 450$ under the Hamerly scaling law. This converts the splitting ratio problem from an architectural constraint to a manufacturing quality control problem.

6. Sub-millisecond computation: The diagnostic runs at approximately 1,860 components per second on standard computing hardware, compatible with wafer-level screening, in-situ monitoring, and real-time mesh reconfiguration.

7. No training data or simulation required: The method operates directly from the constraint equation and the operating point measurement. It requires no electromagnetic simulation, no machine learning training, no internal optical probes, and no iterative optimisation.

8. Cross-domain validated: The diagnostic is validated against unrelated physical systems (quantum Rabi oscillations, thermodynamic critical points), confirming it measures genuine constraint geometry rather than photonic-specific artifacts.

9. Eliminates architectural redundancy: By enabling per-component characterisation, the method removes the need for 3-splitter fault-tolerant MZI designs, recovering the 1.5–2× component overhead and enabling known-good-die screening to replace statistical yield management.

9 RELATIONSHIP TO PARENT APPLICATIONS

This application is a specific industrial application of the methods described in the parent applications:

First parent application (AU 2026902758): Provides the foundational Directive-Based Products framework, the coupling operation, the transposition engine, the invariant closure evaluation, and the adaptive two-channel coupling kernel. The present invention uses the DBP triple formulation to model the MZI as a three-component system (splitting ratio, phase, transmission).

Second parent application (AU 2026902926): Provides the intrinsic geometric diagnostic method (Gaussian curvature via bordered Hessian), the coupling class taxonomy (Class 1 / Class 2), the recursive closure diagnostic, the centrifuge configuration validation, and the cross-domain equivalence detection method. The present invention applies Method 1 (intrinsic geometric diagnostics) to the photonic constraint surface, uses the coupling class taxonomy to establish the exact isolation invariance, applies Method 2 (recursive closure) to characterise complex photonic presets, and uses Method 4 (cross-domain equivalence) for validation.

The present application extends the parent applications by:

(a) Identifying the $K_G = 0$ symmetry condition specific to the 50:50 beam splitter design point;

(b) Proving the exact invariance of K_G under output perturbation as a consequence of the Class 1 structure;

(c) Developing the isolation protocol for cascaded mesh testing;

(d) Connecting the detection threshold to the Hamerly scaling law for quantified manufacturing impact;

(e) Characterising five photonic preset constraint surfaces across the full range of coupler physics;

(f) Establishing the first geometric diagnostic benchmark for silicon photonics.

10 INFORMAL CLAIMS

Note: Formal claims will be drafted for the complete application. The following informal claims indicate the intended scope of protection.

1. A method for localizing splitting ratio faults in individual Mach-Zehnder interferometers within a photonic computing mesh, comprising: (a) for each MZI in the mesh, computing the Gaussian curvature K_G of the coupled-mode theory constraint surface at the MZI's operating point; (b) comparing the computed K_G against a healthy baseline derived from the K_G = 0 symmetry condition at the design splitting ratio; (c) identifying an MZI as faulty when its K_G deviation from baseline exceeds a threshold determined by the measurement noise floor.

2. The method of claim 1, wherein the Gaussian curvature is computed via the bordered Hessian determinant of the coupled-mode constraint equation divided by the fourth power of the gradient magnitude.

3. The method of claim 1, wherein the healthy baseline is derived from a symmetry property of the MZI constraint equation: the vanishing of the derivative of the splitting function $4r(1-r)$ at $r = 0.5$, which renders K_G = 0 as the unique geometric signature of a perfect 50:50 beam splitter.

4. The method of claim 1, wherein the diagnostic is exactly invariant under perturbations of the measured output power due to the linear participation of the output variable in the constraint equation, such that imperfect isolation of adjacent MZIs during testing does not affect the computed K_G.

5. The method of claim 1, further comprising an isolation protocol in which, for each MZI under test, all other MZIs in the mesh are set to a pass-through configuration while the test MZI operates at a diagnostic phase, and the cross-port power transfer is measured to determine the operating point for K_G computation.

6. The method of claim 1, wherein the detection threshold corresponds to a splitting ratio error at which individual component errors begin to dominate the mesh error budget under a scaling law relating residual matrix error to mesh dimension and splitting ratio variance.

7. The method of claim 1, applied to screening photonic computing mesh components during manufacturing, wherein the K_G diagnostic is used to implement known-good-die selection by accepting components with K_G within the healthy baseline tolerance and rejecting components with K_G exceeding the fault threshold.

8. The method of claim 1, wherein the K_G diagnostic is applied to the general coupled-mode theory constraint surface $F(\kappa, \Delta\beta, R) = 0$ to detect gap errors, cladding voids, thermal drift, and combined faults in directional couplers, each fault type producing a distinct K_G signature.

9. The method of claim 1, wherein the K_G diagnostic is invariant under all six permutations of the variable role assignment (which variable is treated as directive, substrate, and product), such that different test station configurations produce the same diagnostic result.

10. The method of claim 1, further comprising cross-domain validation by comparing the photonic constraint surface K_G profile against K_G profiles from unrelated physical systems using the cross-domain equivalence method, to confirm the diagnostic measures genuine constraint geometry.

11. A system for localizing splitting ratio faults in a photonic computing mesh, comprising: a photonic test interface configured to sequentially isolate individual MZIs; a computational processor configured to compute the Gaussian curvature of the coupled-mode constraint surface for each MZI; and a diagnostic output module configured to report faulty components and their inferred splitting ratio errors.

12. The system of claim 11, configured for continuous monitoring during mesh operation, wherein the K_G diagnostic is periodically computed during maintenance windows to detect progressive splitting ratio drift.

13. The system of claim 11, integrated with mesh reconfiguration protocols, wherein detection of a component fault triggers re-routing of computational paths around the degraded component.

14. A computer-readable medium containing instructions that, when executed by a processor, cause the processor to perform the method of any of claims 1 to 10.

END OF PROVISIONAL SPECIFICATION